

## Preparation list (combined)

## 2018 JC2 Prelim Paper 4

| No. | Apparatus                                                                                                                       | Qty                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1   | Goggles                                                                                                                         | 1                   |
| 2   | <b>Dry</b> weighing bottle (new bottle for each shift)                                                                          | 1                   |
| 3   | Styrofoam cup; <b>350</b> cm <sup>3</sup> capacity                                                                              | 2                   |
| 4   | lid for Styrofoam cup , with a central hole into which the thermometer may be inserted                                          | 1                   |
| 5   | thermometer (range –10 °C to +50 °C, 0.2 °C division)                                                                           | 1                   |
| 6   | test–tube rack                                                                                                                  | 1                   |
| 7   | test–tube (big)                                                                                                                 | 6                   |
| 8   | test–tube holder                                                                                                                | 1                   |
| 9   | 250 cm <sup>3</sup> beaker (1 for Qn1, 1 for Qn 2 & 3)                                                                          | 2                   |
| 10  | 50 cm <sup>3</sup> measuring cylinder                                                                                           | 2                   |
| 11  | 25.0 cm <sup>3</sup> pipette                                                                                                    | 1                   |
| 12  | Pipette filler                                                                                                                  | 1                   |
| 13  | 250 cm <sup>3</sup> graduated flask                                                                                             | 1                   |
| 14  | 50 cm <sup>3</sup> burette                                                                                                      | 2                   |
| 15  | Burette clips                                                                                                                   | 2                   |
| 16  | Burette stand and clamp                                                                                                         | 1                   |
| 17  | 250 cm <sup>3</sup> conical flasks                                                                                              | 2                   |
| 18  | White tile                                                                                                                      | 1                   |
| 19  | Glass stirrer                                                                                                                   | 1                   |
| 20  | Glass funnels                                                                                                                   | 3                   |
| 21  | Marker pen                                                                                                                      | 1                   |
| 22  | Wash bottle containing deionised water                                                                                          | 1                   |
| 23  | lighter                                                                                                                         | 1                   |
| 24  | Weighing balances                                                                                                               | 4 per lab           |
| 25  | Airpot (for hot water bath)                                                                                                     | 2 per lab           |
| 26  | 'container for chemical waste' – for organic waste                                                                              | 1 per bench         |
|     | <b>Following items in a plastic bag</b>                                                                                         |                     |
| 27  | Spatula                                                                                                                         | 1                   |
| 28  | Dropping pipette (Qn1 (1), Qn 2 (3), Qn3 (4))                                                                                   | 8                   |
| 29  | Paper towels                                                                                                                    | some                |
| 30  | Wooden splint                                                                                                                   | some                |
| 31  | Litmus paper                                                                                                                    | some                |
| 32  | Filter paper                                                                                                                    | 1                   |
|     | <b>Chemicals</b>                                                                                                                |                     |
| 1   | <b>FA 1</b> is 0.40 mol dm <sup>-3</sup> sulfuric acid, H <sub>2</sub> SO <sub>4</sub> .                                        | 200 cm <sup>3</sup> |
| 2   | <b>FA 2</b> is 0.55 mol dm <sup>-3</sup> sodium hydroxide, NaOH.                                                                | 250 cm <sup>3</sup> |
| 3   | <b>FA 3</b> is Fresh anhydrous sodium carbonate, Na <sub>2</sub> CO <sub>3</sub> . Placed in a <b>dry, stoppered</b> test-tube. | 12 g                |
| 4   | <b>FA 5</b> is 2.00 mol dm <sup>-3</sup> hydrochloric acid, HCl                                                                 | 120 cm <sup>3</sup> |
| 5   | <b>FA 6</b> is 1.00 mol dm <sup>-3</sup> sodium hydroxide, NaOH                                                                 | 150 cm <sup>3</sup> |
| 6   | <b>FA 7</b> is 1.00 mol dm <sup>-3</sup> sodium carbonate, Na <sub>2</sub> CO <sub>3</sub>                                      | 150 cm <sup>3</sup> |
| 7   | <b>FA 8:</b> ethanol, 10% (v/v)                                                                                                 | 10 cm <sup>3</sup>  |
| 8   | <b>FA 9:</b> propanone, 10% (v/v)                                                                                               | 10 cm <sup>3</sup>  |
| 9   | <b>FA 10:</b> ethanoic acid 10% (v/v)                                                                                           | 10 cm <sup>3</sup>  |
| 10  | <b>FA 11:</b> glucose solution 10% (v/v) and ethanal solution 10% (v/v)                                                         | 10 cm <sup>3</sup>  |
|     |                                                                                                                                 |                     |

|    |                                    |   |
|----|------------------------------------|---|
|    | <b>Common bench reagents</b>       |   |
| 11 | Methyl orange indicator            | 1 |
| 12 | aqueous NaOH                       | 1 |
| 13 | Iodine solution                    | 1 |
| 14 | Fehling's solution                 | 1 |
| 15 | Magnesium powder                   | 1 |
| 16 | 2,4–dinitrophenylhydrazine reagent | 1 |
| 17 | Nitric acid                        | 1 |
| 18 | Silver nitrate                     | 1 |
|    |                                    |   |

There are no chemicals **FA 3** and **FA 10**. Students are not required to do the experiments.